

How Prometheus Helps to Monitor a Kubernetes Deployment

The ubiquitous usage of Kubernetes necessitates accurate and timely monitoring of its clusters. Prometheus and Sensu are the tools to latch on to for this.



Development combined with operations leads to the high impact DevOps practices, and microservices based architecture is ubiquitous in such an environment. Although such architecture is not new — it's been around since the 1980s — the DevOps practice is relatively new. The idea of DevOps began in 2008 with a discussion between Patrick Debois and Andrew Clay Shafer, concerning the concept of agile infrastructure. In June 2009, the epochal presentation of '10+ Deploys a Day: Dev and Ops

Cooperation' was made at Flickr by John Allspaw and Paul Hammond. That year can be treated as the year the DevOps movement began. The Linux containerisation technology based Docker came into play in 2013. In 2014, we saw the journey of Kubernetes begin as an orchestration attempt of multiple Docker containers from Google. Subsequently, Kubernetes began to be maintained by the Cloud Native Computing Foundation (CNCF). The consequent proliferation of Kubernetes led to the absolute necessity of monitoring its deployment

in DevOps. This resulted in open source tools like Prometheus and Sensu, along with the older Nagios. Let's take a closer look at Prometheus, while getting the basics on Sensu.

Monitoring tools

We typically talk about the following three tools in the context of Kubernetes monitoring.

1. **Prometheus:** This open source Kubernetes monitoring tool collects the metrics of Kubernetes deployment from various components, relying on the pull

